

THE PROBLEM

7–8%

of global CO₂ emissions come from steelmaking alone.

~60%

of a stainless steel product's footprint is Scope 3 — driven by alloy sourcing.

0

simple tools exist today for plants to simulate “what-if” decarbonization scenarios.

OUR 4-TERM MODEL

Total CO₂ = Raw Materials + Process Route + Energy/Grid +
Alloy Additions

↑ *our differentiator* — Ferrochrome · Ferronickel · NPI

Sources: worldsteel · CEA India · ISSF · IRENA

Most calculators stop at a 3-term model. **Ours adds a 4th: Alloy Additions** —
~60% of a stainless steel's footprint per ISSF.

KEY INSIGHT

Real gap, real numbers: Outokumpu (high-scrap route) ≈ **1.8t** CO₂ vs. EU average **2.8t** — the gap is almost entirely nickel sourcing.

Built for JSL's roadmap: designed to help JSL identify the fastest path to its Net Zero by 2050 commitment.

37.8%

CO₂ REDUCTION

304 Grade steel · same order, two production strategies

3.15t → **1.96t** CO₂ / tonne

LIVE DASHBOARD SNAPSHOT



SCENARIO COMPARISON



Try it live — scan or visit:

jslcc1-five.vercel.app

Sources: worldsteel · CEA India · ISSF · IRENA

V2 Roadmap: Scope 1/2/3 breakdown · Cost overlay · PDF export · Sensitivity analysis